

Full solution packet for arXiv:1612.01364

LRP/ANEP is equivalent to NEP

Status. full_solution_likely_valid. This packet answers the converse question in the remark after Proposition 4.4 of Antonin Dalet and Gilles Lancien, *Some properties of coarse Lipschitz maps between Banach spaces*, arXiv:1612.01364.

Source question. The source defines a net in a metric space as an (a, b) -net with $0 < a \leq b$: distinct net points are at distance at least a , and every ambient point is within distance b of the net. In Section 4 the source defines the net extension property (NEP), proves in Proposition 4.4 that NEP implies the Lipschitz representation property (LRP), and then asks whether the converse is true. The same remark records that LRP is equivalent to an almost net extension property (ANEP).

Proposition 4.4. *Assume that X and Y are Banach spaces such that (X, Y) has the net extension property. Then there exists $\mu \geq 1$ such that (X, Y) has the μ -LRP.*

Proof. Let $f \in \mathcal{CL}(X, Y)$ such that $Lip_\infty(f) < c$. Pick $\mathcal{M}$ be a net in X and $\lambda \geq 1$ such that the conclusion of Lemma 4.2 is satisfied. It follows from an easy change of variable argument that for any $A \geq 1$, the net $A\mathcal{M}$ also satisfies the conclusion of Lemma 4.2 with the same constant λ . Then for A large enough, the restriction of f to $A\mathcal{M}$ is c -Lipschitz. So it admits an extension $g : X \rightarrow Y$ such that g is λc -Lipschitz. Since f and g are both coarse Lipschitz and coincide on a net, it is not difficult to see that $f - g$ is bounded. By adding a constant to g , we may also assume that $g(0) = 0$, which concludes the proof. $\square$

Remark. We do not know if the converse of this last proposition is true. However, it is not difficult to check that the existence of $\mu \geq 1$ such that (X, Y) has the μ -LRP is equivalent to the existence of $\lambda \geq 1$ such that (X, Y) has the λ -ANEP. Here, λ -ANEP stands for λ -almost net extension property, which is formally weaker than NEP and has the following ad'hoc meaning: there exists a net $\mathcal{M}$ in X such that for any Lipschitz function $f : \mathcal{M} \rightarrow Y$, there exists $g : X \rightarrow Y$ Lipschitz so that $Lip(g) \leq \lambda Lip(f)$ and $f - g$ is bounded on $\mathcal{M}$.

Result. The converse is true. For Banach spaces X and Y , NEP, LRP for some constant, and ANEP for some constant are equivalent.

Lemma (Bounded data on a separated set). *Let X be a metric space, let Y be a Banach space, and let $\mathcal{M} \subset X$ be a -separated for some $a > 0$. Every bounded map $e : \mathcal{M} \rightarrow Y$ has a Lipschitz extension $E : X \rightarrow Y$.*

Proof. Let $B = \sup_{m \in \mathcal{M}} \|e(m)\|$. If $B = 0$, take $E = 0$. Otherwise put $r = a/4$ and, for $m \in \mathcal{M}$,

$$\beta_m(x) = \max\{0, 1 - d(x, m)/r\}.$$

The positive supports of the functions β_m are pairwise disjoint. Define

$$E(x) = \begin{cases} \beta_m(x)e(m), & \text{if } \beta_m(x) > 0 \text{ for the unique such } m, \\ 0, & \text{if } \beta_m(x) = 0 \text{ for all } m \in \mathcal{M}. \end{cases}$$

Then $E(m) = e(m)$ for every $m \in \mathcal{M}$. We show that E is Lipschitz. If x, y lie in the same bump support, then

$$\|E(x) - E(y)\| \leq (B/r)d(x, y) = (4B/a)d(x, y).$$

If x lies in the support of β_m and y lies outside that same support with $E(y) = 0$, then $d(y, m) \geq r$, so

$$\|E(x) - E(y)\| \leq B(1 - d(x, m)/r) = B(r - d(x, m))/r \leq (B/r)d(x, y).$$

Finally, if x and y lie in two different bump supports, then

$$d(x, y) \geq a - r - r = a/2,$$

while $\|E(x) - E(y)\| \leq 2B$. Hence $\|E(x) - E(y)\| \leq (4B/a)d(x, y)$ in this case as well. Thus E is Lipschitz. $\square$

Theorem. *Let X and Y be Banach spaces. Then the following are equivalent.*

- (i) (X, Y) has NEP.
- (ii) (X, Y) has μ -LRP for some $\mu \geq 1$.
- (iii) (X, Y) has λ -ANEP for some $\lambda \geq 1$.

Proof. The implication (i) $\Rightarrow$ (ii) is exactly Proposition 4.4 of Dalet–Lancien. The source remark after that proposition records the equivalence (ii) $\Leftrightarrow$ (iii). It remains only to prove (iii) $\Rightarrow$ (i).

Assume that (X, Y) has λ -ANEP, witnessed by a net $\mathcal{M} \subset X$. By the source definition of net, $\mathcal{M}$ is an (a, b) -net for some $0 < a \leq b$, hence it is a -separated. Let $f : \mathcal{M} \rightarrow Y$ be Lipschitz. ANEP gives a Lipschitz map $g : X \rightarrow Y$ such that $f - g$ is bounded on $\mathcal{M}$. Define

$$e(m) = f(m) - g(m) \quad (m \in \mathcal{M}).$$

By the lemma, e has a Lipschitz extension $E : X \rightarrow Y$. Then

$$F = g + E$$

is Lipschitz on X , and for every $m \in \mathcal{M}$,

$$F(m) = g(m) + e(m) = f(m).$$

Thus every Lipschitz map $\mathcal{M} \rightarrow Y$ admits a Lipschitz extension to X . So $(\mathcal{M}, X, Y)$ has the Lipschitz extension property, and therefore (X, Y) has NEP. $\square$

Consequences. This proves the converse of Proposition 4.4 in the source paper. In particular, the “almost” bounded-error formulation is not weaker than NEP for the separated nets used by Dalet–Lancien: bounded errors can be removed by local Lipschitz bumps.

Limitations. This packet does not settle the later remark asking whether $(CL(X, Y), \|\cdot\|_{CL})$ is always complete. Proposition 4.5 says that LRP, hence equivalently NEP, is sufficient for completeness, but the source does not state that it is necessary.

Verification notes. The source definition of `net`, Proposition 4.4, and the following remark were checked directly. The run indexes were searched for the exact arXiv id and for `ANEP`, `almost net extension`, `LRP`, and `net extension property`; no existing packet for this converse target was found. A previous packet for the same arXiv id proves only a finite-dimensional-domain positive case for the later completeness question.

Human-review recommendation. Review as a short full solution of the converse/LRP-vs-NEP question. The only substantive point is the elementary bounded-data extension lemma; the separation constant $a > 0$ in the source definition of `net` is essential.

References

- [1] A. Dalet and G. Lancien, *Some properties of coarse Lipschitz maps between Banach spaces*, arXiv:1612.01364.